

Synchronization of stochastic coupled systems via feedback control based on discrete-time state observations



Yongbao Wu, Bingdao Chen, Wenxue Li *

Department of Mathematics, Harbin Institute of Technology (Weihai), Weihai, 264209, PR China

ARTICLE INFO

Article history:

Received 15 September 2016

Accepted 25 April 2017

Available online 26 May 2017

Keywords:

Synchronization

Stochastic coupled systems

Discrete-time state

Feedback control

Graph theory

ABSTRACT

In this paper, we formulate and investigate the synchronization of stochastic coupled systems via feedback control based on discrete-time state observations (SCSFD). The discrete-time state feedback control is used in the drift parts of response system. Combining Lyapunov method with graph theory, the upper bound of duration between two consecutive state observations is provided. And a global Lyapunov function of SCSFD is presented, which derives some sufficient criteria to guarantee the synchronization of drive–response systems in the sense of mean-square asymptotical synchronization. In addition, the theoretical results are applied to stochastic coupled oscillators and second-order Kuramoto oscillators. Finally, two numerical examples are given to verify the effectiveness of the theoretical results.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

In recent years, considerable attention has been devoted to the study of coupled systems due to their extensive applications in many fields including mathematical [1,2], physics [3,4], biology [5–7]. Therein, it is noteworthy that a large quantity of coupled systems are discussed in a determinate case in relevant literatures [2,3,8,9]. However, in real applications, dynamical behaviors of coupled systems are affected inevitably by stochastic interference which exists in external environment. Hence, it is necessary to take random disturbance into account in the research of coupled systems. In fact, there are many papers related to stochastic coupled systems which mainly investigated the dynamic properties of stochastic coupled systems such as stability [1,10–12], synchronization [9,13–17], oscillatory property [18] and other dynamic properties [19,20].

As one of most important dynamical behaviors, synchronization indicates that the dynamical behaviors of coupled systems achieve the same time spatial state. For example, when electronic systems are synchronized, events at points far apart appear simultaneous or near-simultaneous from a certain perspective [21]. The investigation on the synchronization not only shows the principle of the synchronization but also can obtain the theoretical bases to take advantage of this character for benefiting human beings. Synchronization has many practical applications in real life. For instance, Hu et al. [22] proposed a regional traffic signal synchronization strategy to alleviate traffic congestion and to improve traffic efficiency in the network. Karimi-Ghartemani et al. [23] presented a new synchronization method which employs an enhanced phase-locked loop system to solve the problem of power electronic converters. Pursley et al. [24] investigated the synchronization of ovulation in dairy cows using PGF2alpha and GnRH. Above facts in life reveal the ubiquity of synchronization. Hence, it is necessary to study the synchronization of stochastic coupled systems.

* Corresponding author. Fax: +86 0631 5687660.

E-mail address: wenxue810823@163.com (W. Li).

In order to make the stochastic coupled systems synchronized, we should impose controllers on them in most cases. In applications, there are many types of concepts of control such as adaptive control [25–27], optimal control [28,29], fuzzy control [30], observer-based control [31], feedback control [32–36] and so on. Among these controls, feedback control as a simple strategy can be effectively applied to synchronizing stochastic coupled systems due to its reasonable process of taking control by comparing the input value with output value of systems. There are two kinds of observation states in which feedback control is based on, continuous-time state and discrete-time state. We find that most existing literature of feedback control are based on continuous-time state [32–34]. However, discrete-time state observations are more practical than continuous-time state observations. For example, relevant data is collected by meteorological station through discrete-time state observation in weather prediction. Compared with data which comes from continuous-time, these data can also lead to a good result, moreover, meteorological station hardware requirement is low. Therefore, it is more efficient and costs less in practice for observing state only at discrete times. On account of these merits, investigating discrete-time state observations is meaningful and several results related to this issue are provided in [36–41].

Recently, there have been many results on synchronization and feedback control based on discrete-time state observations in literature. For instance, in [9], the reliable phase synchronization problem between two coupled chaotic fractional order systems was studied by constructing an active nonlinear feedback control scheme. A new approach to synchronization analysis of linearly coupled ordinary differential systems was derived based on geometrical analysis of the synchronization manifold by Lu et al. in [13]. Mao et al. [37] discussed the mean-square exponential stabilization of continuous-time hybrid stochastic differential equations by feedback controls based on discrete-time state observations. In [38], Pages et al. solved nonlinear filtering problems associated with discrete-time or continuous-time state processes and discrete-time observations by an optimal quantization approach. From the examples cited above, we learn that synchronization is practical and the feedback control based on discrete-time state observations has great potentialities. However, to the best of our knowledge, there is no result in literature investigating synchronization of stochastic coupled systems via feedback control based on discrete-time state observations (SCSFD). Hence, we will try to study the synchronization of SCSFD.

In the automatic control theory, Lyapunov method is a powerful tool to study synchronization. However, it is a challenging work to construct an appropriate Lyapunov function for stochastic systems. In [1], Li et al. developed a systematic approach that allows one to construct global Lyapunov functions for large-scale coupled systems by building blocks of individual vertex systems to solve this problem. Motivated by above analysis, we will study the synchronization of SCSFD by using Lyapunov method and graph theory in this paper. To ensure drive–response systems in the sense of mean-square asymptotical synchronization, we establish some sufficient conditions. After that, we apply theoretic results to stochastic coupled oscillators and second-order Kuramoto oscillators. Finally, two numerical examples will be given. The main contributions of the paper are highlighted as follows:

1. We use feedback control based on discrete-time state observations to investigate synchronization of stochastic coupled systems and get the upper bound of the duration between two consecutive state observations.

2. We construct global Lyapunov functions for SCSFD via building vertex-Lyapunov function of each vertex system, which avoids finding an appropriate Lyapunov function directly. Moreover, some sufficient criteria are obtained for the drive–response systems in the sense of mean-square asymptotical synchronization.

The remainder of this paper is outlined as follows. In Section 2, we present a mathematical model of SCSFD and give some preliminaries. In Section 3, some sufficient criteria for the sense of mean-square asymptotical synchronization of drive–response systems are provided. In Section 4, we employ theoretical results in stochastic coupled oscillators and second-order Kuramoto oscillators. Finally, two numerical examples are provided in Section 5.

2. Preliminaries and model formulation

2.1. Preliminaries

We introduce some notations firstly. Let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a complete probability space with a filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$ satisfying the usual conditions (i.e., it is right continuous and $\mathcal{F}_0$ contains all $\mathbb{P}$ -null sets). And $B(t)$ is a one-dimensional Brownian motion defined on the probability space. The mathematical expectation with respect to the given probability measure $\mathbb{P}$ is denoted by $\mathbb{E}(\cdot)$ and $\mathbb{R}^n$ stands for n -dimensional Euclidean space. The superscript “T” stands for the transpose of a vector or matrix. If $x \in \mathbb{R}^n$, then $|x|$ is Euclidean norm. The notations $\mathbb{L} = \{1, 2, \dots, N\}$, $\mathbb{Z}^+ = \{1, 2, \dots\}$, $\mathbb{R}^+ = [0, +\infty)$ and $m = \sum_{k=1}^N m_k$ for $m_k \in \mathbb{Z}^+$ are used. The $C^{2,1}(\mathbb{R}^n \times \mathbb{R}^+; \mathbb{R}^+)$ represents the family of all nonnegative functions $V(x, t)$ on $\mathbb{R}^n \times \mathbb{R}^+$ which are continuously twice differentiable in x and once in t . A digraph $\mathcal{G}$ is weighted if each arc (i, j) is assigned a positive weight a_{ij} and we call $A = (a_{ij})_{N \times N}$ as the weight matrix. And P is the Laplacian matrix of $(\mathcal{G}, A)$, in which P is defined as $P = (p_{kh})_{N \times N}$, where $p_{kh} = -a_{kh}$ for $k \neq h$ and $p_{kh} = \sum_{j \neq k} a_{kj}$ for $k = h$.

A useful lemma, which can be easily found in literature [1], is as follows:

Lemma 1 ([1]). Suppose that $N \geq 2$. Let $c^{(k)}$ denote the cofactor of the k th diagonal element of P which stands for the Laplacian matrix of $(\mathcal{G}, A)$. Then the following identity holds:

$$\sum_{k,h=1}^N c^{(k)} a_{kh} F_{kh}(x_k, x_h) = \sum_{Q \in \mathcal{Q}} W(Q) \sum_{(s,r) \in E(C_Q)} F_{rs}(x_r, x_s),$$

where for any $k, h \in \mathbb{L}$, $F_{kh}(x_k, x_h) : \mathbb{R}^{m_k} \times \mathbb{R}^{m_h} \rightarrow \mathbb{R}$ is an arbitrary function, $\mathcal{Q}$ is the set of all spanning unicyclic graphs of digraph $(\mathcal{G}, A)$ that denotes the digraph with weight matrix A , $W(\mathcal{Q})$ is the weight of $\mathcal{Q}$, and $\mathcal{C}_{\mathcal{Q}}$ denotes the directed cycle of $\mathcal{Q}$. Particularly, if digraph $(\mathcal{G}, A)$ is strongly connected, then $c^{(k)} > 0$ for any $k \in \mathbb{L}$.

2.2. Model formulation

To begin with, we establish a drive system defined on a digraph $\mathcal{G}$ with N ($N \geq 2$) vertices in the following:

$$dx_k(t) = \left(f_k(x_k(t), t) + \sum_{h=1}^N a_{kh} H_{kh}(x_k(t), x_h(t)) \right) dt + g_k(x_k(t), t) dB(t), t \geq 0, k \in \mathbb{L}, \quad (1)$$

where $x_k(t) \in \mathbb{R}^{m_k}$ is the state vector of the k th vertex system, $f_k, g_k : \mathbb{R}^{m_k} \times \mathbb{R}^+ \rightarrow \mathbb{R}^{m_k}$ are continuous functions, $H_{kh} : \mathbb{R}^{m_k} \times \mathbb{R}^{m_h} \rightarrow \mathbb{R}^{m_k}$ represents coupling form and satisfies global Lipschitz condition for any x_k, x_h , and a_{kh} stands for coupling strength.

In order to synchronize drive–response systems, we design a discrete-time state observations feedback control which can be described as $u_k(y_k(\delta_t) - x_k(\delta_t), t) : \mathbb{R}^{m_k} \times \mathbb{R}^+ \rightarrow \mathbb{R}^{m_k}$. Then, the response system with discrete-time state observations feedback control can be written as follows:

$$dy_k(t) = \left(f_k(y_k(t), t) + u_k(y_k(\delta_t) - x_k(\delta_t), t) + \sum_{h=1}^N a_{kh} H_{kh}(y_k(t), y_h(t)) \right) dt + g_k(y_k(t), t) dB(t), t \geq 0, k \in \mathbb{L}, \quad (2)$$

where $\delta_t = [t/\sigma_k]\sigma_k$, $\sigma_k > 0$ is the duration between two consecutive state observations, and $[t/\sigma_k]$ is the integer of t/σ_k . In order to better understand system (2), we make the following diagram (see Fig. 1).

Further, let $e_k(t) = y_k(t) - x_k(t)$ be the k th synchronization error state vector between drive–response systems (1) and (2). Therefore, the synchronization error system is described by

$$de_k(t) = \left(f_k(y_k(t), t) - f_k(x_k(t), t) + u_k(e_k(\delta_t), t) + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(t), y_h(t)) - H_{kh}(x_k(t), x_h(t))) \right) dt + (g_k(y_k(t), t) - g_k(x_k(t), t)) dB(t), t \geq 0, k \in \mathbb{L}, \quad (3)$$

where $e_k = (e_k^{(1)}, e_k^{(2)}, \dots, e_k^{(m_k)})^T$.

Next, we will introduce three definitions about differential operator, vertex-Lyapunov function and synchronicity in mean square.

Definition 1. Let function $V^k(e_k, t) \in C^{2,1}(\mathbb{R}^{m_k} \times \mathbb{R}^+; \mathbb{R}^+)$. Differential operator $LV^k(e_k, t)$ is defined by

$$\begin{aligned} LV^k(e_k, t) &= V_t^k(e_k, t) + V_e^k(e_k, t) \left[f_k(y_k(t), t) - f_k(x_k(t), t) + u_k(e_k(\delta_t), t) + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(t), y_h(t)) - H_{kh}(x_k(t), x_h(t))) \right] \\ &\quad + \frac{1}{2} \text{trace} \left[(g_k(y_k(t), t) - g_k(x_k(t), t))^T V_{ee}^k(e_k, t) (g_k(y_k(t), t) - g_k(x_k(t), t)) \right], \end{aligned}$$

in which

$$V_t^k(e_k, t) = \frac{\partial V^k(e_k, t)}{\partial t}, V_e^k(e_k, t) = \left(\frac{\partial V^k(e_k, t)}{\partial e_k^{(1)}}, \dots, \frac{\partial V^k(e_k, t)}{\partial e_k^{(m_k)}} \right) \text{ and } V_{ee}^k(e_k, t) = \left(\frac{\partial^2 V^k(e_k, t)}{\partial e_k^{(i)} \partial e_k^{(j)}} \right)_{m_k \times m_k}.$$

Definition 2. Function $V^k(e_k, t) \in C^{2,1}(\mathbb{R}^{m_k} \times \mathbb{R}^+; \mathbb{R}^+)$ is called vertex-Lyapunov function for system (3) if following conditions are satisfied for any $k \in \mathbb{L}$.

A1. There is a positive constant ζ_k satisfying

$$\mathbb{E}LV^k(e_k, t) \leq -\zeta_k \mathbb{E}|e_k|^2 + \mathbb{E} \sum_{h=1}^N a_{kh} F_{kh}(e_k, e_h).$$

A2. For each directed cycle $\mathcal{C}_{\mathcal{Q}}$ of weighted digraph $(\mathcal{G}, A)$, there is

$$\sum_{(h,k) \in E(\mathcal{C}_{\mathcal{Q}})} F_{kh}(e_k, e_h) \leq 0, \quad (4)$$

for any $e_k \in \mathbb{R}^{m_k}$, $e_h \in \mathbb{R}^{m_h}$.

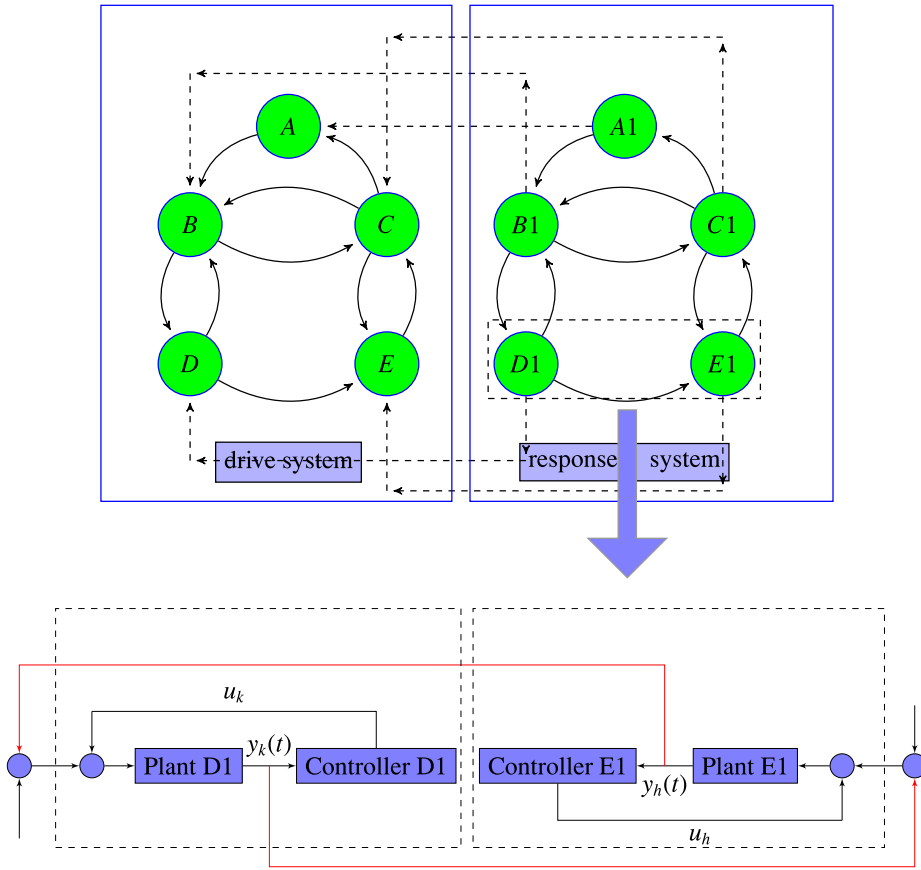


Fig. 1. Drive–response systems and the model for a response system.

Definition 3 ([42]). The drive system (1) and response system (2) are said to be synchronous in mean square if the error system (3) is stable in mean square, that is, for any $x_0 \in \mathbb{R}^m$,

$$\lim_{T \rightarrow \infty} \mathbb{E} \int_0^T |e(t, x_0)|^2 dt < \infty. \quad (5)$$

3. Stabilization analysis for SCSFD

We obtain some forms of sufficient criteria to assure the synchronization of drive–response systems (1) and (2) in the sense of mean-square asymptotical synchronization by Lyapunov method and graph theory in this section. Suppose that digraph $(\mathcal{G}, A)$ is always strongly connected. In this section, some assumptions and the definition of mean-square asymptotical synchronization are presented.

Assumption 1. There exist positive constants α_k and β_k such that

$$|f_k(y_k, t) - f_k(x_k, t)| \leq \alpha_k |e_k| \quad \text{and} \quad |g_k(y_k, t) - g_k(x_k, t)| \leq \beta_k |e_k|, \quad \forall k \in \mathbb{L}, \quad (6)$$

for any $(x_k, y_k, e_k, t) \in \mathbb{R}^{m_k} \times \mathbb{R}^{m_k} \times \mathbb{R}^{m_k} \times \mathbb{R}^+$.

Assumption 2. Suppose that $u_k(e_k, t)$ satisfies the global Lipschitz condition for any $k \in \mathbb{L}$, in other words, there exists a positive constant ν_k which follows the condition as

$$|u_k(e_k, t) - u_k(\bar{e}_k, t)| \leq \nu_k |e_k - \bar{e}_k|, \quad \forall t \in \mathbb{R}^+, \quad (7)$$

for any $(e_k, \bar{e}_k) \in \mathbb{R}^{m_k} \times \mathbb{R}^{m_k}$. Besides,

$$u_k(0, t) = 0, \quad \forall t \in \mathbb{R}^+.$$

Assumption 3. Assume that $H_{kh}(x_k, x_h)$ satisfies the locally Lipschitz condition and there are positive constants o_k and ι_k , such that

$$|H_{kh}(y_k, y_h) - H_{kh}(x_k, x_h)| \leq o_k |e_k| + \iota_k |e_h|, \quad (8)$$

for any $(x_k, x_h), (y_k, y_h) \in \mathbb{R}^{m_k} \times \mathbb{R}^{m_h}$.

Remark 1. We assume $u_k(x_k, t), f_k(x_k, t), g_k(x_k, t)$ and $H_{kh}(x_k, x_h)$ satisfy linear growth condition. Combined with Assumptions 1–3 and Theorem 3.1 of second Chapter in literature [43], we can know that system (3) has a unique solution $e(t) = (e_1^T(t), e_2^T(t), \dots, e_N^T(t))^T$ with initial value $e(0) = e_0$, where $e_0 \in \mathbb{R}^m$, which means system (3) has a trivial solution $e(t) \equiv 0$.

Definition 4. The drive–response systems (1) and (2) are said to be mean-square asymptotically synchronized if $\lim_{t \rightarrow \infty} \mathbb{E}|e(t)|^2 = 0$, for all initial value $e_0 \in \mathbb{R}^m$.

In the following subsections, a theoretical framework for constructing a Lyapunov function of system (3) will be established. Now, we provide a Lyapunov-type theorem as follows.

3.1. Lyapunov-type theorem

Theorem 1. Suppose digraph $(\mathcal{G}, A)$ is strongly connected where $A = (a_{kh})_{N \times N}$. If system (3) admits vertex-Lyapunov functions $V^k(e_k, t)$, then the drive–response systems (1) and (2) are synchronization in mean square.

Proof. Define Lyapunov function

$$V(e, t) = \sum_{k=1}^N c^{(k)} V^k(e_k, t),$$

where $c^{(k)}$ represents the cofactor of the k th diagonal element of P . Since digraph $(\mathcal{G}, A)$ is strongly connected, we can obtain $c^{(k)} > 0$, for any $k \in \mathbb{L}$ by Lemma 1. According to condition A1 we can get

$$\begin{aligned} \mathbb{E}LV(e, t) &= \mathbb{E} \sum_{k=1}^N c^{(k)} LV^k(e_k, t) \\ &\leq \sum_{k=1}^N c^{(k)} \left[-\zeta_k \mathbb{E}|e_k|^2 + \mathbb{E} \sum_{h=1}^N a_{kh} F_{kh}(e_k, e_h) \right] \\ &= -\sum_{k=1}^N c^{(k)} \zeta_k \mathbb{E}|e_k|^2 + \mathbb{E} \sum_{k,h=1}^N c^{(k)} a_{kh} F_{kh}(e_k, e_h). \end{aligned}$$

By using Lemma 1, we have

$$\sum_{k,h=1}^N c^{(k)} a_{kh} F_{kh}(e_k, e_h) = \sum_{\mathcal{Q} \in \mathcal{Q}} W(\mathcal{Q}) \sum_{(h,k) \in E(\mathcal{C}_{\mathcal{Q}})} F_{kh}(e_k, e_h).$$

In view of condition A2, it is easy to get

$$\mathbb{E}LV(e, t) \leq -\sum_{k=1}^N c^{(k)} \zeta_k \mathbb{E}|e_k|^2. \quad (9)$$

For any fixed $e_0 \in \mathbb{R}^m$, applying the Itô's formula and (9), we can show that

$$\begin{aligned} \mathbb{E}V(e(t), t) &= V(e_0, 0) + \mathbb{E} \int_0^t LV(e(s), s) ds \\ &\leq C - \int_0^t \sum_{k=1}^N c^{(k)} \zeta_k \mathbb{E}|e_k(s)|^2 ds \\ &\leq C - B \int_0^t \mathbb{E} \sum_{k=1}^N |e_k(s)|^2 ds \\ &= C - B \int_0^t \mathbb{E}|e(s)|^2 ds, \end{aligned} \quad (10)$$

in which $B = \min_{k \in \mathbb{L}} \{c^{(k)} \zeta_k\}$, $C = V(e_0, 0)$ and C is a positive constant. Through (10), we can derive that

$$\int_0^t \mathbb{E}|e(s)|^2 ds \leq \frac{C}{B}. \quad (11)$$

Thus, According to Definition 3, this result is derived. $\square$

Remark 2. In Theorem 1, we assume digraph $(\mathcal{G}, A)$ is strongly connected and construct a global Lyapunov function, which enables the drive–response systems (1) and (2) are synchronization in mean square. In the process of proving Theorem 1, besides direct calculation, we have another way to verify the condition A2. If there exist functions $P_k(e_k) : \mathbb{R}^{m_k} \rightarrow \mathbb{R}$, for any $k \in \mathbb{L}$, satisfying

$$F_{kh}(e_k, e_h) \leq P_k(e_k) - P_h(e_h).$$

In that way the condition of Theorem 1 is verified easily. For example, in Fig. 2, we have

$$\begin{aligned} \sum_{(h,k) \in E(\mathcal{C}_{\mathcal{Q}})} F_{kh}(e_k, e_h) &\leq \sum_{(h,k) \in E(\mathcal{C}_{\mathcal{Q}})} P_k(e_k) - P_h(e_h) \\ &= P_1(e_1) - P_2(e_2) + P_2(e_2) - P_4(e_4) + P_4(e_4) - P_3(e_3) + P_3(e_3) - P_1(e_1) \\ &\quad + P_2(e_2) - P_1(e_1) + P_1(e_1) - P_3(e_3) + P_3(e_3) - P_4(e_4) + P_4(e_4) - P_2(e_2) \\ &= 0. \end{aligned}$$

Remark 3. The feature of this paper is that the stochastic coupled systems are described by a digraph. With the help of graph theory, stochastic coupled systems can be studied much easily. It is well-known that the Lyapunov method is an important approach for investigating the synchronization of systems. From Theorem 1, we get a systematic method to construct a Lyapunov function for system (3) by using Lyapunov functions V^k of each vertex system, which avoids looking for a Lyapunov function directly.

The criteria above are based on the Lyapunov function. Besides, we will give a coefficient-type criterion based on Theorem 1 in the sense of synchronization in mean square of drive–response systems (1) and (2), which makes conditions in the form of coefficients easier to be checked.

3.2. Coefficient-type criterion

Theorem 2. Under Assumptions 1–3, if digraph $(\mathcal{G}, A)$ is strongly connected and the following conditions are fulfilled, then the synchronization of drive–response systems (1) and (2) are synchronization in mean square, for all initial value $e_0 \in \mathbb{R}^m$ and any $k \in \mathbb{L}$.

B1. There is a positive constant ε_k such that

$$e_k^T [f_k(y_k, t) - f_k(x_k, t) + u_k(e_k, t)] \leq -\varepsilon_k |e_k|^2. \quad (12)$$

B2. There is a positive constant ϵ_k such that

$$2\varepsilon_k > 4\epsilon_k + \beta_k^2 + 2o_k \sum_{h=1}^N a_{kh} + 2t_k \sum_{h=1}^N a_{kh} + \lambda_k, \quad (13)$$

where

$$\lambda_k = \frac{3v_k^2 \sigma_k}{2\epsilon_k} [3\sigma_k(\alpha_k^2 + 2v_k^2) + \beta_k^2] + \frac{9v_k^2 \sigma_k^2}{\epsilon_k} \left(\left(\sum_{h=1}^N a_{kh} o_k \right)^2 + l_k^2 N \sum_{h=1}^N a_{kh}^2 \right) \quad \text{and} \quad \sigma_k < \frac{1}{3\sqrt{2}v_k}.$$

Proof. Define functions

$$\begin{aligned} V^k(e_k, t) &= |e_k|^2 + \vartheta_k \int_{t-\sigma_k}^t \int_s^t \left[\sigma_k |f_k(y_k(v), v) - f_k(x_k(v), v) + u_k(e_k(\delta_v), v) \right. \\ &\quad \left. + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(v), y_h(v)) - H_{kh}(x_k(v), x_h(v)))|^2 + |g_k(y_k(v), v) - g_k(x_k(v), v)|^2 \right] dv ds, \end{aligned}$$

where ϑ_k is a positive constant which will be determined. Using the Itô's formula and conditions (6–8) and (12), it follows that

$$\begin{aligned}
& LV^k(e_k(t), t) \\
&= 2e_k^T(t) \left[f_k(y_k(t), t) - f_k(x_k(t), t) + u_k(e_k(\delta_t), t) + \sum_{h=1}^N a_{kh}(H_{kh}(y_k(t), y_h(t)) - H_{kh}(x_k(t), x_h(t))) \right] \\
&\quad + \text{trace} \left[(g_k(y_k(t), t) - g_k(x_k(t), t))^T (g_k(y_k(t), t) - g_k(x_k(t), t)) \right] \\
&\quad + \vartheta_k \sigma_k \left[\sigma_k |f_k(y_k(t), t) - f_k(x_k(t), t) + u_k(e_k(\delta_t), t)|^2 \right. \\
&\quad \left. + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(t), y_h(t)) - H_{kh}(x_k(t), x_h(t)))^2 + |g_k(y_k(t), t) - g_k(x_k(t), t)|^2 \right] \\
&\quad - \vartheta_k \int_{t-\sigma_k}^t \left[\sigma_k |f_k(y_k(s), s) - f_k(x_k(s), s) + u_k(e_k(\delta_s), s)|^2 \right. \\
&\quad \left. + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(s), y_h(s)) - H_{kh}(x_k(s), x_h(s)))^2 + |g_k(y_k(s), s) - g_k(x_k(s), s)|^2 \right] ds \\
&\leq (-2\varepsilon_k + \beta_k^2) |e_k(t)|^2 + 4\epsilon_k |e_k(t)|^2 + \frac{\nu_k^2}{4\epsilon_k} |e_k(t) - e_k(\delta_t)|^2 + 2 \sum_{h=1}^N a_{kh} |e_k(t)| [o_k |e_k(t)| + \iota_k |e_h(t)|] \\
&\quad + \vartheta_k \sigma_k \left[\sigma_k |f_k(y_k(t), t) - f_k(x_k(t), t) + u_k(e_k(\delta_t), t)|^2 \right. \\
&\quad \left. + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(t), y_h(t)) - H_{kh}(x_k(t), x_h(t)))^2 + |g_k(y_k(t), t) - g_k(x_k(t), t)|^2 \right] \\
&\quad - \vartheta_k \int_{t-\sigma_k}^t \left[\sigma_k |f_k(y_k(s), s) - f_k(x_k(s), s) + u_k(e_k(\delta_s), s)|^2 \right. \\
&\quad \left. + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(s), y_h(s)) - H_{kh}(x_k(s), x_h(s)))^2 + |g_k(y_k(s), s) - g_k(x_k(s), s)|^2 \right] ds. \tag{14}
\end{aligned}$$

Applying the elementary inequality, we gain

$$2 \sum_{h=1}^N a_{kh} |e_k(t)| [o_k |e_k(t)| + \iota_k |e_h(t)|] \leq 2(o_k + \iota_k) \sum_{h=1}^N a_{kh} |e_k(t)|^2 + \iota_k \sum_{h=1}^N a_{kh} (|e_h(t)|^2 - |e_k(t)|^2). \tag{15}$$

By using conditions (6–8), we can get that

$$\begin{aligned}
& \vartheta_k \sigma_k \left[\sigma_k |f_k(y_k(t), t) - f_k(x_k(t), t) + u_k(e_k(\delta_t), t)|^2 \right. \\
&\quad \left. + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(t), y_h(t)) - H_{kh}(x_k(t), x_h(t)))^2 + |g_k(y_k(t), t) - g_k(x_k(t), t)|^2 \right] \\
&\leq \vartheta_k \sigma_k \left[3\sigma_k \left(|f_k(y_k(t), t) - f_k(x_k(t), t)|^2 + |u_k(e_k(\delta_t), t)|^2 \right) \right. \\
&\quad \left. + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(t), y_h(t)) - H_{kh}(x_k(t), x_h(t)))^2 + |g_k(y_k(t), t) - g_k(x_k(t), t)|^2 \right] \\
&\leq \vartheta_k \sigma_k [3\sigma_k (\alpha_k^2 |e_k(t)|^2 + \nu_k^2 |e_k(\delta_t)|^2) + \beta_k^2 |e_k(t)|^2] \\
&\quad + 3\vartheta_k \sigma_k^2 \sum_{h=1}^N a_{kh} (H_{kh}(y_k(t), y_h(t)) - H_{kh}(x_k(t), x_h(t)))^2
\end{aligned}$$

$$\begin{aligned}
&\leq \vartheta_k \sigma_k \left[3\sigma_k (\alpha_k^2 + 2\nu_k^2) + \beta_k^2 \right] |e_k(t)|^2 + 6\vartheta_k \sigma_k^2 \nu_k^2 |e_k(t) - e_k(\delta_t)|^2 \\
&\quad + 6\vartheta_k \sigma_k^2 \left(\left(\sum_{h=1}^N a_{kh} o_k \right)^2 + \iota_k^2 N \sum_{h=1}^N a_{kh}^2 \right) |e_k(t)|^2 + 6\vartheta_k \sigma_k^2 \iota_k^2 N \sum_{h=1}^N a_{kh}^2 (|e_h(t)|^2 - |e_k(t)|^2) \\
&\leq d_k |e_k(t)|^2 + 6\vartheta_k \sigma_k^2 \nu_k^2 |e_k(t) - e_k(\delta_t)|^2 + 6\vartheta_k \sigma_k^2 \iota_k^2 N \sum_{h=1}^N a_{kh}^2 (|e_h(t)|^2 - |e_k(t)|^2),
\end{aligned} \tag{16}$$

where

$$d_k = \vartheta_k \sigma_k \left[3\sigma_k (\alpha_k^2 + 2\nu_k^2) + \beta_k^2 \right] + 6\vartheta_k \sigma_k^2 \left(\left(\sum_{h=1}^N a_{kh} o_k \right)^2 + \iota_k^2 N \sum_{h=1}^N a_{kh}^2 \right).$$

Substituting (15) and (16) into (14), we can conduct the following calculation

$$\begin{aligned}
&LV^k(e_k(t), t) \\
&\leq -\kappa_k |e_k(t)|^2 + \left(\frac{\nu_k^2}{4\epsilon_k} + 6\vartheta_k \sigma_k^2 \nu_k^2 \right) |e_k(t) - e_k(\delta_t)|^2 + \sum_{h=1}^N a_{kh} (\iota_k + 6\vartheta_k \sigma_k^2 \iota_k^2 N a_{kh}) (|e_h(t)|^2 - |e_k(t)|^2) \\
&\quad - \vartheta_k \int_{t-\sigma_k}^t \left[\sigma_k |f_k(y_k(s), s) - f_k(x_k(s), s) + u_k(e_k(\delta_s), s)| \right. \\
&\quad \left. + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(s), y_h(s)) - H_{kh}(x_k(s), x_h(s)))^2 + |g_k(y_k(s), s) - g_k(x_k(s), s)|^2 \right] ds,
\end{aligned} \tag{17}$$

in which $\kappa_k = 2\epsilon_k - 4\epsilon_k - \beta_k^2 - 2o_k \sum_{h=1}^N a_{kh} - 2\iota_k \sum_{h=1}^N a_{kh} - d_k$. By calculating the expectation of both sides of inequality (17), it holds that

$$\begin{aligned}
&\mathbb{E}LV^k(e_k(t), t) \\
&\leq -\kappa_k \mathbb{E}|e_k(t)|^2 + \left(\frac{\nu_k^2}{4\epsilon_k} + 6\vartheta_k \sigma_k^2 \nu_k^2 \right) \mathbb{E}|e_k(t) - e_k(\delta_t)|^2 \\
&\quad + \mathbb{E} \sum_{h=1}^N a_{kh} (\iota_k + 6\vartheta_k \sigma_k^2 \iota_k^2 N a_{kh}) (|e_h(t)|^2 - |e_k(t)|^2) \\
&\quad - \vartheta_k \mathbb{E} \int_{t-\sigma_k}^t \left[\sigma_k |f_k(y_k(s), s) - f_k(x_k(s), s) + u_k(e_k(\delta_s), s)| \right. \\
&\quad \left. + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(s), y_h(s)) - H_{kh}(x_k(s), x_h(s)))^2 + |g_k(y_k(s), s) - g_k(x_k(s), s)|^2 \right] ds.
\end{aligned} \tag{18}$$

Noting that $t \leq \delta_t + \sigma_k$ for all $t > 0$, according to system (3) we have

$$\begin{aligned}
&\mathbb{E}|e_k(t) - e_k(\delta_t)|^2 \\
&\leq 2\mathbb{E} \int_{t-\sigma_k}^t \left[\sigma_k |f_k(y_k(s), s) - f_k(x_k(s), s) + u_k(e_k(\delta_s), s)| \right. \\
&\quad \left. + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(s), y_h(s)) - H_{kh}(x_k(s), x_h(s)))^2 + |g_k(y_k(s), s) - g_k(x_k(s), s)|^2 \right] ds.
\end{aligned} \tag{19}$$

Particularly, let $\vartheta_k = \frac{3\nu_k^2}{2\epsilon_k}$ and $\sigma_k < \frac{1}{3\sqrt{2}\nu_k}$. Thus, we can replace d_k with λ_k , where

$$\lambda_k = \frac{3\nu_k^2 \sigma_k}{2\epsilon_k} \left[3\sigma_k (\alpha_k^2 + 2\nu_k^2) + \beta_k^2 \right] + 9 \frac{\nu_k^2 \sigma_k^2}{\epsilon_k} \left(\left(\sum_{h=1}^N a_{kh} o_k \right)^2 + \iota_k^2 N \sum_{h=1}^N a_{kh}^2 \right).$$

Then, combining (18) with (19), it is easy to derive that

$$\mathbb{E}LV^k(e_k(t), t) \leq -\kappa_k \mathbb{E}|e_k(t)|^2 + \mathbb{E} \sum_{h=1}^N a_{kh} \left(\iota_k + 6 \frac{v_k^2}{\epsilon_k} \sigma_k^2 \iota_k^2 N a_{kh} \right) (|e_h(t)|^2 - |e_k(t)|^2).$$

According to Theorem 1 and Remark 2. We know $\int_0^\infty \mathbb{E}|e(t)|^2 dt < \infty$. Hence, the proof of theorem is complete. $\square$

Remark 4. It is not difficult to see that the upper bound of σ_k which is provided from condition B2 of Theorem 2 is closely related with Lipschitz constant v_k of control function u_k .

Remark 5. In general, we know that from $\int_0^\infty \mathbb{E}|e(t)|^2 dt < \infty$ cannot drive that $\lim_{t \rightarrow \infty} \mathbb{E}|e(t)|^2 = 0$. However, in this paper, we will give the following theorem to illustrate that is possible.

Theorem 3. The drive–response systems (1) and (2) are asymptotically synchronized in mean square if all conditions of Theorem 2 are satisfied.

Proof. On account of $e(t) = (e_1^T(t), e_2^T(t), \dots, e_N^T(t))^T$, it is easy to see that

$$\mathbb{E}|e(t)|^2 = \sum_{k=1}^N \mathbb{E}|e_k(t)|^2.$$

For any $k \in \mathbb{L}$, fixed any $e_k(0) \in \mathbb{R}^{m_k}$. By using the Itô's formula, we can derive that

$$\begin{aligned} \mathbb{E}|e_k(t)|^2 &= |e_k(0)|^2 + \mathbb{E} \int_0^t \left(2e_k^T(v) \left[f_k(y_k(v), v) - f_k(x_k(v), v) + u_k(e_k(\delta_v), v) \right. \right. \\ &\quad \left. \left. + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(v), y_h(v)) - H_{kh}(x_k(v), x_h(v))) \right] + |g_k(y_k(v), v) - g_k(x_k(v), v)|^2 \right) dv, \forall t \geq 0. \end{aligned}$$

From Assumptions 1–3, it shows that

$$\mathbb{E}|e_k(t)|^2 \leq |e_k(0)|^2 + D \int_0^t \mathbb{E}|e_k(v)|^2 dv + D \int_0^t \mathbb{E}|e_k(v) - e_k(\delta_v)|^2 dv + D \int_0^t \mathbb{E}|e(v)|^2 dv, \quad (20)$$

where D which stands for a positive constant with no actual meaning always changes. So we write all constants as D . Now, for any integer $s \geq 0$, $v \in [s\sigma_k, (s+1)\sigma_k]$, let $\delta_z = s\sigma_k$, $z \in [s\sigma_k, v]$, we can conclude it from (3) that

$$\begin{aligned} e_k(v) - e_k(\delta_v) &= \int_{\delta_v}^v \left[f_k(y_k(z), z) - f_k(x_k(z), z) + u_k(e_k(\delta_z), z) + \sum_{h=1}^N a_{kh} (H_{kh}(y_k(z), y_h(z)) - H_{kh}(x_k(z), x_h(z))) \right] dz \\ &\quad + \int_{\delta_v}^v (g_k(y_k(z), z) - g_k(x_k(z), z)) dB(z). \end{aligned}$$

Using Assumptions 1–3 and Cauchy–Schwarz inequality, it follows that

$$\begin{aligned} \mathbb{E}|e_k(v) - e_k(\delta_v)|^2 &\leq \left(6\sigma_k \alpha_k^2 + 12\sigma_k \delta_k^2 \left(\sum_{h=1}^N a_{kh} \right)^2 + 2\beta_k^2 \right) \int_{\delta_v}^v \mathbb{E}|e_k(z)|^2 dz \\ &\quad + 12\sigma_k N \iota_k^2 \max_{h \in \mathbb{L}} \{a_{kh}^2\} \int_{\delta_v}^v \mathbb{E}|e(z)|^2 dz \\ &\quad + 12\sigma_k^2 v_k^2 (\mathbb{E}|e_k(v)|^2 + \mathbb{E}|e_k(v) - e_k(\delta_v)|^2). \end{aligned}$$

Because $\sigma_k \leq \frac{1}{3\sqrt{2}v_k}$, it is easy to see that $12\sigma_k^2 v_k^2 < 1$, then we can get

$$\mathbb{E}|e_k(v) - e_k(\delta_v)|^2 \leq D \int_{\delta_v}^v \mathbb{E}|e_k(z)|^2 dz + D \int_{\delta_v}^v \mathbb{E}|e(z)|^2 dz + \frac{12\sigma_k^2 v_k^2}{1 - 12\sigma_k^2 v_k^2} \mathbb{E}|e_k(v)|^2. \quad (21)$$

By substituting (21) into (20), we can obtain that

$$\begin{aligned} \mathbb{E}|e_k(t)|^2 &\leq |e_k(0)|^2 + D \int_0^t \mathbb{E}|e_k(v)|^2 dv + D \int_0^t \mathbb{E}|e(v)|^2 dv \\ &\quad + D \int_0^t \int_{\delta_v}^v \mathbb{E}|e_k(z)|^2 dz dv + D \int_0^t \int_{\delta_v}^v \mathbb{E}|e(z)|^2 dz dv. \end{aligned} \quad (22)$$

In view of $v - \sigma_k \leq \delta_v$, it yields that

$$\begin{aligned} \int_0^t \int_{\delta_v}^v \mathbb{E}|e_k(z)|^2 dz dv &\leq \int_0^t \int_{v-\sigma_k}^v \mathbb{E}|e_k(z)|^2 dz dv \\ &\leq \int_{-\sigma_k}^t \mathbb{E}|e_k(z)|^2 \int_z^{z+\sigma_k} dv dz \\ &\leq \sigma_k \int_{-\sigma_k}^t \mathbb{E}|e_k(z)|^2 dz. \end{aligned} \quad (23)$$

Similarly, we obtain

$$\int_0^t \int_{\delta_v}^v \mathbb{E}|e(z)|^2 dz dv \leq \sigma_k \int_{-\sigma_k}^t \mathbb{E}|e(z)|^2 dz. \quad (24)$$

Substituting (23) and (24) into (22), and with the help of Theorem 2, it is obvious that

$$\mathbb{E}|e_k(t)|^2 \leq D, \forall t \geq 0. \quad (25)$$

Using Itô's formula, we can conclude the following calculation

$$\begin{aligned} &\mathbb{E}|e_k(t_2)|^2 - \mathbb{E}|e_k(t_1)|^2 \\ &= \mathbb{E} \int_{t_1}^{t_2} \left(2e_k^\top(t) \left[f_k(y_k(t), t) - f_k(x_k(t), t) + u_k(e_k(\delta_t), t) \right. \right. \\ &\quad \left. \left. + \sum_{k=1}^N a_{kh} (H_{kh}(y_k(t), y_h(t)) - H_{kh}(x_k(t), x_h(t))) \right] + |g_k(y_k(t), t) - g_k(x_k(t), t)|^2 \right) dt, \end{aligned}$$

for any $0 \leq t_1 < t_2 < \infty$. According to (25) and Assumptions 1–3, we derive that

$$\mathbb{E}|e_k(t_2)|^2 - \mathbb{E}|e_k(t_1)|^2 \leq D(t_2 - t_1).$$

Moreover, we have

$$\mathbb{E}|e(t_2)|^2 - \mathbb{E}|e(t_1)|^2 = \sum_{k=1}^N (\mathbb{E}|e_k(t_2)|^2 - \mathbb{E}|e_k(t_1)|^2) \leq D(t_2 - t_1).$$

Therefore, we prove that $\mathbb{E}|e(t)|^2$ is uniformly continuous in $0 \leq t < \infty$. It satisfies the condition that $\lim_{t \rightarrow \infty} \mathbb{E}|e(t)|^2 = 0$, that is, the drive–response systems (1) and (2) are asymptotically synchronized in mean square. This completes the proof. $\square$

4. An application to stochastic coupled oscillators

We will apply the main results to discuss the mean-square asymptotical synchronization of stochastic coupled oscillators in this section. Consider the following nonlinear oscillators:

$$\ddot{x}(t) + \phi \dot{x}(t) + \varrho x(t) = 0, t \geq 0,$$

in which $\phi \geq 0$ is a damping coefficient. Then, the synchronization of stochastic coupled oscillators will be investigated. These stochastic coupled oscillators are modeled on a digraph $\mathcal{G}$ which has N vertices. To begin with, stochastic coupled oscillators are described by

$$\ddot{x}_k(t) + \phi_k \dot{x}_k(t) + \varrho_k x_k(t) + \sum_{h=1}^N a_{kh} H_{kh}(x_k(t), x_h(t)) = h_k(x_k(t), t) \dot{B}(t), t \geq 0, k \in \mathbb{L}, \quad (26)$$

where a_{kh} stands for the intensity of influence from vertex h to vertex k and $\varrho_k > 0$. We can obtain that $a_{kh} \geq 0$, and $a_{kh} = 0$ if and only if there exists no arc from vertex h to vertex k in $\mathcal{G}$. Meanwhile, coupling form and the intensity of perturbation existing in the k th vertex are described as $H_{kh} : \mathbb{R}^1 \times \mathbb{R}^1 \rightarrow \mathbb{R}^1$ and $h_k : \mathbb{R}^1 \times \mathbb{R}^+ \rightarrow \mathbb{R}^1$. Then, we make a transformation of $\tilde{x}_k(t) = \dot{x}_k(t) + \xi_k x_k(t)$, and get the system as follows:

$$\begin{cases} d\tilde{x}_k(t) = [\tilde{x}_k(t) - \xi_k x_k(t)] dt, \\ d\tilde{x}_k(t) = [(-\phi_k + \xi_k) \tilde{x}_k(t) + (\xi_k \phi_k - \xi_k^2 - \varrho_k) x_k(t)] dt \\ \quad - \sum_{h=1}^N a_{kh} H_{kh}(x_k(t), x_h(t)) dt + h_k(x_k(t), t) dB(t), \end{cases} \quad t \geq 0, k \in \mathbb{L}. \quad (27)$$

We consider the system (27) as a drive system. Accordingly, the response system in which we insert a discrete-time state feedback control is written as follows:

$$\begin{cases} dy_k(t) = [\tilde{y}_k(t) - \xi_k y_k(t) + a_k y_k(\delta_t) - x_k(\delta_t)] dt, \\ d\tilde{y}_k(t) = [(-\phi_k + \xi_k) \tilde{y}_k(t) + (\xi_k \phi_k - \xi_k^2 - \varrho_k) y_k(t) + b_k(\tilde{y}_k(\delta_t) - \tilde{x}_k(\delta_t))] dt \\ \quad - \sum_{h=1}^N a_{kh} H_{kh}(y_k(t), y_h(t)) dt + h_k(y_k(t), t) dB(t), \end{cases} \quad t \geq 0, k \in \mathbb{L}. \quad (28)$$

For the sake of simplification, let $X_k(t) = (x_k(t), \tilde{x}_k(t))^T$, $Y_k(t) = (y_k(t), \tilde{y}_k(t))^T$, $E_k(t) = (y_k(t) - x_k(t), \tilde{y}_k(t) - \tilde{x}_k(t))^T$,

$$F_k(X_k(t), t) = \begin{pmatrix} \tilde{x}_k(t) - \xi_k x_k(t) \\ (-\phi_k + \xi_k) \tilde{x}_k(t) + (\xi_k \phi_k - \xi_k^2 - \varrho_k) x_k(t) \end{pmatrix}$$

and

$$F_k(Y_k(t), t) = \begin{pmatrix} \tilde{y}_k(t) - \xi_k y_k(t) \\ (-\phi_k + \xi_k) \tilde{y}_k(t) + (\xi_k \phi_k - \xi_k^2 - \varrho_k) y_k(t) \end{pmatrix}.$$

Therefore, we can rewrite system (27) and (28) as

$$dX_k(t) = \left[F_k(X_k(t), t) + \sum_{h=1}^N a_{kh} \tilde{H}_{kh}(X_k(t), X_h(t)) \right] dt + \tilde{h}_k(X_k(t), t) dB(t), k \in \mathbb{L} \quad (29)$$

and

$$dY_k(t) = \left[F_k(Y_k(t), t) + \sum_{h=1}^N a_{kh} \tilde{H}_{kh}(Y_k(t), Y_h(t)) + U_k(Y_k(\delta_t) - X_k(\delta_t), t) \right] dt + \tilde{h}_k(Y_k(t), t) dB(t), k \in \mathbb{L}, \quad (30)$$

where

$$\tilde{H}_{kh}(X_k(t), X_h(t)) = \begin{pmatrix} 0 \\ -H_{kh}(x_k(t), x_h(t)) \end{pmatrix}, \tilde{H}_{kh}(Y_k(t), Y_h(t)) = \begin{pmatrix} 0 \\ -H_{kh}(y_k(t), y_h(t)) \end{pmatrix},$$

$$\tilde{h}_k(X_k(t), t) = \begin{pmatrix} 0 \\ h_k(x_k(t), t) \end{pmatrix}, \tilde{h}_k(Y_k(t), t) = \begin{pmatrix} 0 \\ h_k(y_k(t), t) \end{pmatrix}$$

and

$$U_k(Y_k(\delta_t) - X_k(\delta_t), t) = \begin{pmatrix} a_k(y_k(\delta_t) - x_k(\delta_t)) \\ b_k(\tilde{y}_k(\delta_t) - \tilde{x}_k(\delta_t)) \end{pmatrix}.$$

Hence, we can get immediately the following synchronization error system:

$$\begin{aligned} dE_k(t) = & \left[F_k(Y_k(t), t) - F_k(X_k(t), t) + U_k(E_k(\delta_t), t) + \sum_{h=1}^N a_{kh} (\tilde{H}_{kh}(Y_k(t), Y_h(t)) - \tilde{H}_{kh}(X_k(t), X_h(t))) \right] dt \\ & + [\tilde{h}_k(Y_k(t), t) - \tilde{h}_k(X_k(t), t)] dB(t), k \in \mathbb{L}. \end{aligned} \quad (31)$$

Then, we will introduce a theorem about the mean-square asymptotical synchronization of drive–response systems (29) and (30) in the following.

Theorem 4. Assuming the following conditions hold, then the drive–response systems (29) and (30) are mean-square asymptotically synchronized.

C1. There exists positive constant ω_k such that

$$|h_k(y_k, t) - h_k(x_k, t)| \leq \omega_k |e_k|.$$

C2. There exist positive constants ρ_k and η_k such that

$$|H_{kh}(y_k, y_h) - H_{kh}(x_k, x_h)| \leq \rho_k |e_k| + \eta_k |e_h|.$$

C3. There exists positive constant ϵ_k such that

$$2\gamma_k > 4\epsilon_k + \omega_k^2 + 2\rho_k \sum_{h=1}^N a_{kh} + 2\eta_k \sum_{h=1}^N a_{kh} + \lambda_k,$$

in which

$$\lambda_k = \frac{3\nu_k^2 \sigma_k}{2\epsilon_k} [3\sigma_k(\alpha_k^2 + 2\nu_k^2) + \omega_k^2] + 9 \frac{\nu_k^2 \sigma_k^2}{\epsilon_k} \left(\left(\sum_{h=1}^N a_{kh} \rho_k \right)^2 + \eta_k^2 N \sum_{h=1}^N a_{kh}^2 \right) \quad \text{and} \quad \sigma_k < \frac{1}{3\sqrt{2}\nu_k},$$

where $\gamma_k = \min\{\varpi_k, \varsigma_k\}$, $\varpi_k = -\frac{\xi_k\phi_k + |\xi_k^2 + \varrho_k - 1|}{2} - \phi_k - \xi_k - b_k > 0$, $\varsigma_k = -\frac{\xi_k\phi_k + |\xi_k^2 + \varrho_k - 1|}{2} - a_k + \xi_k > 0$, $\nu_k = \sqrt{\max\{a_k^2, b_k^2\}}$ and $\alpha_k^2 = \max\left\{\xi_k(1 + \xi_k) + (\xi_k\phi_k + \xi_k^2 + \varrho_k)^2 + 2\xi_k^2\phi_k + \xi_k^3 + \xi_k\varrho_k + \xi_k\phi_k^2 + \phi_k\varrho_k, 1 + \xi_k + (\xi_k + \phi_k)^2 + 2\xi_k^2\phi_k + \xi_k^3 + \xi_k\varrho_k + \xi_k\phi_k^2 + \phi_k\varrho_k\right\}$.

Proof. Let $e_k(t) = y_k(t) - x_k(t)$ and $\tilde{e}_k(t) = \tilde{y}_k(t) - \tilde{x}_k(t)$. In the view of condition C1, we can get

$$\begin{aligned} & E_k^T(t)[F_k(Y_k(t), t) - F_k(X_k(t), t) + U_k(E_k(t), t)] \\ & \leq |\xi_k\phi_k - \xi_k^2 - \varrho_k + 1||e_k(t)\tilde{e}_k(t)| + (\alpha_k - \xi_k)e_k^2(t) - (\phi_k - \xi_k - b_k)\tilde{e}_k^2(t) \\ & \leq \left[\frac{|\xi_k\phi_k - \xi_k^2 - \varrho_k + 1|}{2} - (\phi_k - \xi_k - b_k)\right]\tilde{e}_k^2(t) + \left[\frac{|\xi_k\phi_k - \xi_k^2 - \varrho_k + 1|}{2} + a_k - \xi_k\right]e_k^2(t) \\ & \leq \left[\frac{\xi_k\phi_k + |\xi_k^2 + \varrho_k - 1|}{2} + \phi_k + \xi_k + b_k\right]\tilde{e}_k^2(t) + \left[\frac{\xi_k\phi_k + |\xi_k^2 + \varrho_k - 1|}{2} + a_k - \xi_k\right]e_k^2(t) \\ & \leq -\gamma_k|E_k(t)|^2, \end{aligned} \quad (32)$$

where $\gamma_k = \min\{\varpi_k, \varsigma_k\}$. It means that condition B1 is satisfied. Then we conclude that

$$\begin{aligned} & |F_k(Y_k(t), t) - F_k(X_k(t), t)|^2 \\ & = |\tilde{e}_k(t) - \xi_k e_k(t)|^2 + |(-\phi_k + \xi_k)\tilde{e}_k(t) + (\xi_k\phi_k - \xi_k^2 - \varrho_k)e_k(t)|^2 \\ & \leq \left[\xi_k(1 + \xi_k) + (\xi_k\phi_k + \xi_k^2 + \varrho_k)^2 + 2\xi_k^2\phi_k + \xi_k^3 + \xi_k\varrho_k + \xi_k\phi_k^2 + \phi_k\varrho_k\right]e_k^2(t) \\ & \quad + [1 + \xi_k + (\xi_k + \phi_k)^2 + 2\xi_k^2\phi_k + \xi_k^3 + \xi_k\varrho_k + \xi_k\phi_k^2 + \phi_k\varrho_k]\tilde{e}_k^2(t) \\ & \leq \alpha_k^2|E_k(t)|^2. \end{aligned} \quad (33)$$

Since $\nu_k = \sqrt{\max\{a_k^2, b_k^2\}}$, we can get

$$|U_k(E_k, t) - U_k(\bar{E}_k, t)| \leq \nu_k|E_k - \bar{E}_k|. \quad (34)$$

It is easy to see that Assumptions 1, 2 and 3 are satisfied according to (32), (33) and (34). From condition C3, we have

$$2\epsilon_k > 4\epsilon_k + \beta_k^2 + 2o_k \sum_{h=1}^N a_{kh} + 2t_k \sum_{h=1}^N a_{kh} + \lambda_k,$$

where

$$\lambda_k = \frac{3\nu_k^2\sigma_k}{2\epsilon_k}[3\sigma_k(\alpha_k^2 + 2\nu_k^2) + \beta_k^2] + 9\frac{\nu_k^2\sigma_k^2}{\epsilon_k}\left(\left(\sum_{h=1}^N a_{kh}o_k\right)^2 + \iota_k^2N \sum_{h=1}^N a_{kh}^2\right) \quad \text{and} \quad \sigma_k < \frac{1}{3\sqrt{2}\nu_k}.$$

This means that condition B2 is satisfied, so all conditions of Theorem 2 are fulfilled. Therefore, we show that the drive-response systems (29) and (30) are synchronization in mean square based on Theorem 2. Also, we can conclude that the drive-response systems (29) and (30) are asymptotically synchronized in mean square according to Theorem 3. This proof is complete. $\square$

5. Numerical test

In this section, we will give two examples with numerical simulations to illustrate the effectiveness of theoretical results obtained in the above section.

Example 1. We consider the stochastic coupled oscillators with discrete-time state feedback control. Model for system (31) is expressed by a digraph $(\mathcal{G}, A)$ with N vertices, where $N = 4$ (see Fig. 2), and weighted matrix is shown as follows:

$$A = (a_{kh})_{4 \times 4} = \begin{pmatrix} 0 & 0.001 & 0.002 & 0 \\ 0.003 & 0 & 0 & 0.004 \\ 0.003 & 0 & 0 & 0.001 \\ 0 & 0.004 & 0.002 & 0 \end{pmatrix}.$$

This digraph $(\mathcal{G}, A)$ is strongly connected. We suppose that $\phi_k = 0.1$, $H_{kh}(x_k, x_h) = 0.2x_h - 0.2x_k$, $h_k(x_k, t) = 0.3x_k$, where $k = 1, 2, 3, 4$. Also, parameters are assumed to be $\xi_1 = 0.001$, $\xi_2 = \xi_3 = 0.003$, $\xi_4 = 0.002$ and $\varrho_1 = \varrho_2 = \varrho_3 = \varrho_4 = 1$.

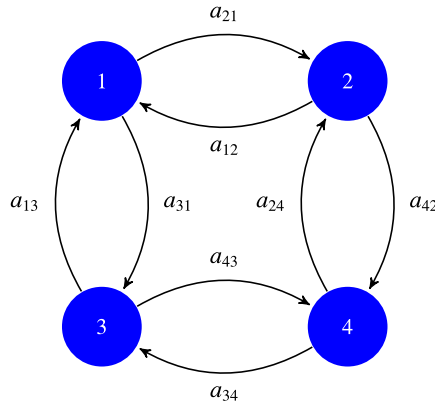


Fig. 2. A strong connected digraph $\mathcal{G}$ with 4 vertices.

Then, we will design a discrete-time state feedback control in order to synchronize drive system (29). The control function is described by

$$U_k(E_k(\delta_t), t) = A_k E_k(\delta_t),$$

where $A_k = \begin{pmatrix} a_k & 0 \\ 0 & b_k \end{pmatrix}$. We choose $a_k = b_k = -0.65$, and let $\epsilon_k = 0.1$, $\sigma_k = (1/8)v_k$. By a simple computation, we can get $\omega_k = 0.3$, $\rho_k = 0.2$, $\eta_k = 0.2$, $v_k = 0.65$, $\sum_{h=1}^4 a_{1h} = 0.003$, $\sum_{h=1}^4 a_{2h} = 0.007$, $\sum_{h=1}^4 a_{3h} = 0.004$, $\sum_{h=1}^4 a_{4h} = 0.006$,

$$\alpha_k^2 = \begin{cases} 1.1122 & k = 1, \\ 1.1166 & k = 2, 3, \\ 1.1144 & k = 4, \end{cases} \gamma_k = \begin{cases} 0.5489 & k = 1, \\ 0.5468 & k = 2, 3, \\ 0.5479 & k = 4 \end{cases} \text{ and } \lambda_k = \begin{cases} 0.291995752 & k = 1, \\ 0.292549210 & k = 2, \\ 0.292548276 & k = 3, \\ 0.292272752 & k = 4. \end{cases}$$

By a simple calculation, we can check all conditions in Theorem 4 with the following condition:

$$\begin{aligned} 2\gamma_1 - 4\epsilon_1 - \omega_1^2 - 2\rho_1 \sum_{h=1}^4 a_{1h} - 2\eta_1 \sum_{h=1}^4 a_{1h} - \lambda_1 &= 0.3134 > 0, \\ 2\gamma_2 - 4\epsilon_2 - \omega_2^2 - 2\rho_2 \sum_{h=1}^4 a_{2h} - 2\eta_2 \sum_{h=1}^4 a_{2h} - \lambda_2 &= 0.3055 > 0, \\ 2\gamma_3 - 4\epsilon_3 - \omega_3^2 - 2\rho_3 \sum_{h=1}^4 a_{3h} - 2\eta_3 \sum_{h=1}^4 a_{3h} - \lambda_3 &= 0.3079 > 0, \\ 2\gamma_4 - 4\epsilon_4 - \omega_4^2 - 2\rho_4 \sum_{h=1}^4 a_{4h} - 2\eta_4 \sum_{h=1}^4 a_{4h} - \lambda_4 &= 0.3092 > 0. \end{aligned}$$

Therefore, we know that the drive–response systems (27) and (28) are asymptotically synchronized in mean square. Also, the mean square figure and the sample path figure of system (29)–(31) are shown in Figs. 3–8, which reveal the feasibility and effectiveness of theoretical results.

Example 2. The synchronization process of coupled oscillators of large populations is a hot topic in various scientific disciplines such as physics, biology, and sociology. As we know, the power network is closely related to the interacted Kuramoto oscillators, in particular, the second-order Kuramoto-like model [44,45]. Moreover, in literature [45], Thomas et al. investigated second-order Kuramoto oscillators as follows:

$$\frac{d^2\theta_i}{dt^2} = -\alpha \frac{d\theta_i}{dt} + \Omega_i + \lambda \sum_{j=1}^N A_{ij} \sin[\theta_j(t) - \theta_i(t)], \quad (35)$$

where α is the dissipating parameter, λ is the coupling strength and Ω_i is the natural frequency of oscillator i . Next, we consider the second-order Kuramoto oscillators with stochastic interference as follows:

$$\ddot{x}_k(t) + \psi \dot{x}_k(t) - \Omega_k - \varphi \sum_{h=1}^N a_{kh} \sin[x_h(t) - x_k(t)] = \beta x_k(t) \dot{B}(t), \quad t \geq 0, \quad k \in \mathbb{L}.$$

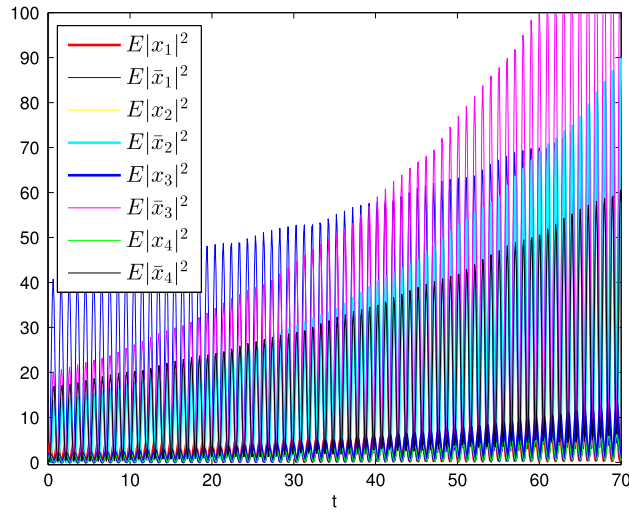


Fig. 3. The mean square figure of system (29) when $N = 4$.

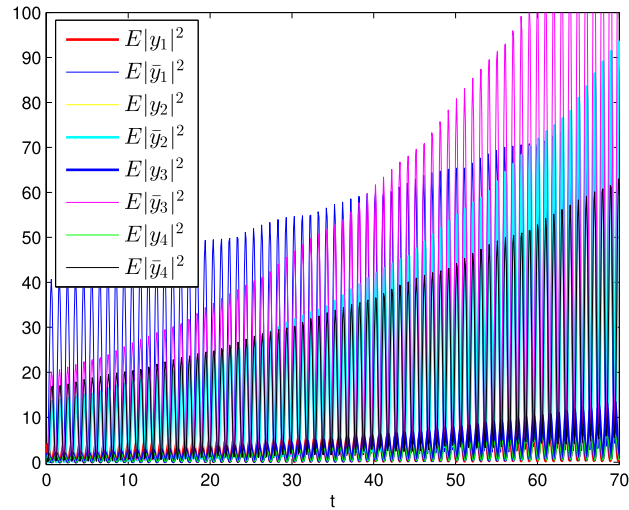


Fig. 4. The mean square figure of system (30) when $N = 4$.

Then, we let $\tilde{x}_k(t) = \dot{x}_k(t) + \xi_k x_k(t)$, drive–response systems have same handle method with drive–response stochastic coupled oscillators system. Therefore, we can obtain the following error system.

$$\begin{cases} d\tilde{e}_k(t) = [\tilde{e}_k(t) - \xi_k e_k(t) + a_k e_k(\delta_t)] dt, \\ d\tilde{e}_k(t) = [(-\psi + \xi_k) \tilde{e}_k(t) + (\xi_k \psi - \xi_k^2) e_k(t) + b_k \tilde{e}_k(\delta_t)] dt \\ - \varphi \sum_{h=1}^N a_{kh} \sin[e_h(t) - e_k(t)] dt + \beta e_k(t) dB(t), \end{cases} \quad t \geq 0, \quad k \in \mathbb{L}. \quad (36)$$

Comparing with the system (31), we have $\phi_k = \psi$, $\varrho_k = 0$, $H_{kh}(x_k, x_h) = \varphi \sin[x_h(t) - x_k(t)]$. And the digraph with weighted matrix $A = (a_{kh})_{N \times N}$ is given as follows:

$$A = \begin{pmatrix} 0 & 0.003 & 0 & 0 & 0.002 \\ 0.001 & 0 & 0.004 & 0 & 0 \\ 0 & 0.002 & 0 & 0.005 & 0 \\ 0 & 0 & 0.001 & 0 & 0.001 \\ 0.004 & 0 & 0 & 0.002 & 0 \end{pmatrix}.$$

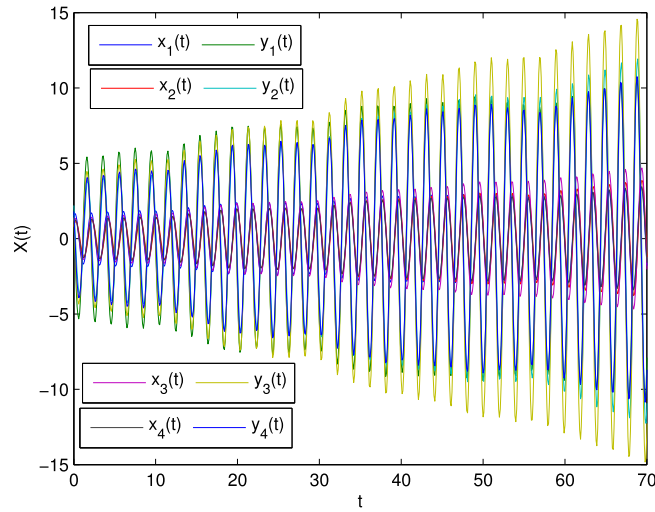


Fig. 5. The sample path figure of system (29) when $N = 4$.

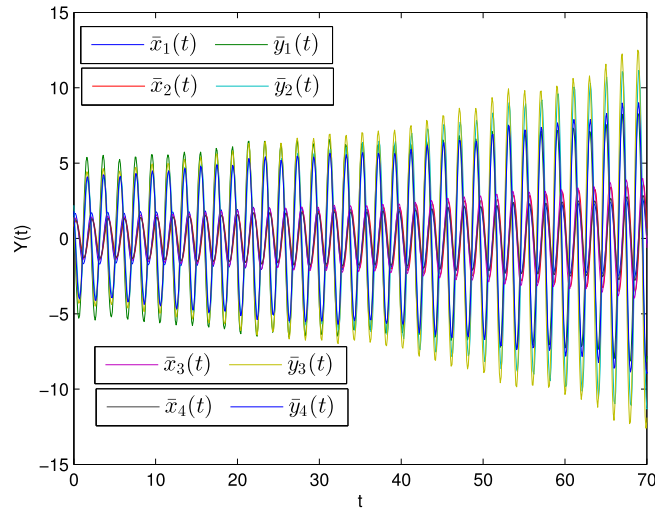


Fig. 6. The sample path figure of system (30) when $N = 4$.

We choose $a_k = b_k = -1$, $\epsilon_k = 0.1$, $\sigma_k = (1/18)v_k$, $\xi_1 = 0.002$, $\xi_2 = 0.003$, $\xi_3 = 0.001$, $\xi_4 = \xi_5 = 0.004$, $\psi = 0.01$, $\varphi = 0.05$, $\beta = 0.1$. By a simple computation, we can get $\omega_k = 0.1$, $\rho_k = 0.05$, $\eta_k = 0.05$, $v_k = 1$, $\sum_{h=1}^5 a_{1h} = 0.005$, $\sum_{h=1}^5 a_{2h} = 0.005$, $\sum_{h=1}^5 a_{3h} = 0.007$, $\sum_{h=1}^5 a_{4h} = 0.002$, $\sum_{h=1}^5 a_{5h} = 0.006$,

$$\alpha_k^2 = \begin{cases} 1.0021 & k = 1, \\ 1.0032 & k = 2, \\ 1.0011 & k = 3, \\ 1.0042 & k = 4, \\ 1.0042 & k = 5, \end{cases} \gamma_k = \begin{cases} 0.4880 & k = 1, \\ 0.4870 & k = 2, \\ 0.4890 & k = 3, \\ 0.4860 & k = 4, \\ 0.4860 & k = 5, \end{cases} \text{ and } \lambda_k = \begin{cases} 0.425291729 & k = 1, \\ 0.425444521 & k = 2, \\ 0.425152912 & k = 3, \\ 0.425583343 & k = 4, \\ 0.425583428 & k = 5. \end{cases}$$

Moreover, we can check all conditions in Theorem 4 with the following conditions:

$$2\gamma_1 - 4\epsilon_1 - \omega_1^2 - 2\rho_1 \sum_{h=1}^5 a_{1h} - 2\eta_1 \sum_{h=1}^5 a_{1h} - \lambda_1 = 0.1397 > 0,$$

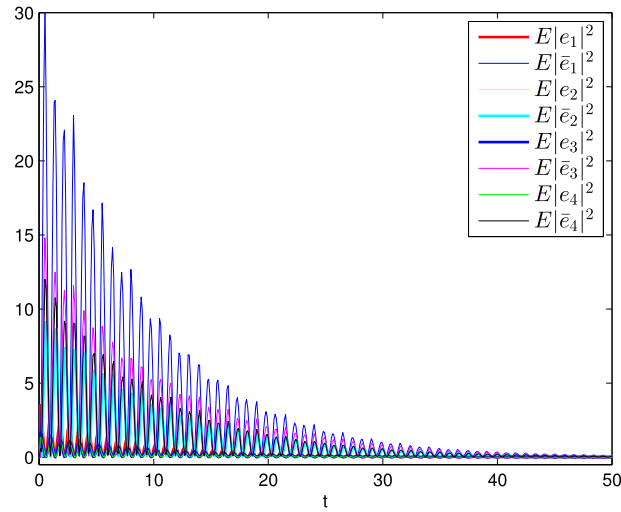


Fig. 7. The mean square figure of system (31) when $N = 4$.

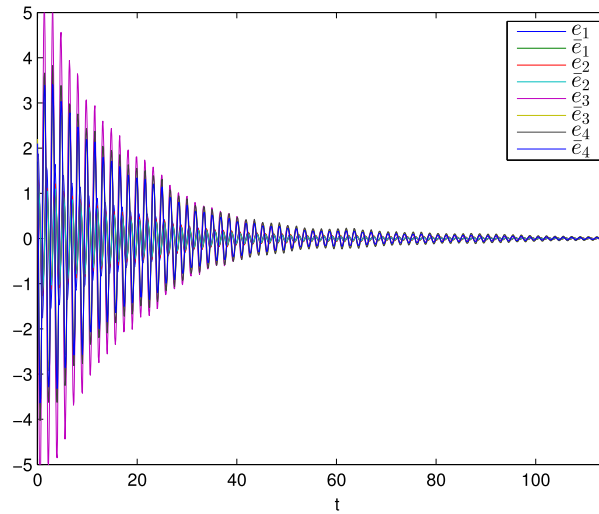


Fig. 8. The sample path figure of system (31) when $N = 4$.

$$2\gamma_2 - 4\epsilon_2 - \omega_2^2 - 2\rho_2 \sum_{h=1}^5 a_{2h} - 2\eta_2 \sum_{h=1}^5 a_{2h} - \lambda_2 = 0.1376 > 0,$$

$$2\gamma_3 - 4\epsilon_3 - \omega_3^2 - 2\rho_3 \sum_{h=1}^5 a_{3h} - 2\eta_3 \sum_{h=1}^5 a_{3h} - \lambda_3 = 0.1414 > 0,$$

$$2\gamma_4 - 4\epsilon_4 - \omega_4^2 - 2\rho_4 \sum_{h=1}^5 a_{4h} - 2\eta_4 \sum_{h=1}^5 a_{4h} - \lambda_4 = 0.1360 > 0,$$

$$2\gamma_5 - 4\epsilon_5 - \omega_5^2 - 2\rho_5 \sum_{h=1}^5 a_{5h} - 2\eta_5 \sum_{h=1}^5 a_{5h} - \lambda_5 = 0.1352 > 0.$$

It means that all conditions in Theorem 4 are verified. Thus, the trivial solution of system (36) is mean-square exponentially stable. The corresponding numerical result is shown in Fig. 9, which shows the effectiveness and feasibility of our theoretical results.

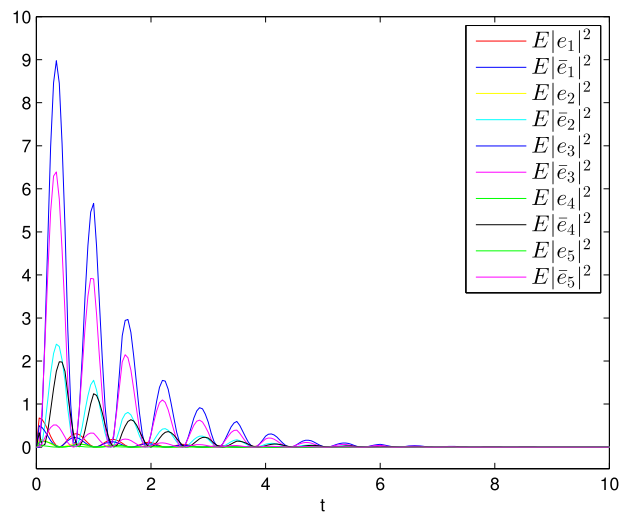


Fig. 9. The mean square of system (36) when $N = 5$.

Acknowledgments

This work was supported by the NNSF of China (Nos. 11301112 and 11401136), the NSF of Shandong Province (Nos. ZR2013AQ003 and ZR2014AQ010), China Postdoctoral Science Foundation funded project (No. 2014T70313) and HIT. IBRSEM. A. 2014014.

References

- [1] M.Y. Li, Z. Shuai, Global-stability problem for coupled systems of differential equations on networks, *J. Differential Equations* 248 (2010) 1–20.
- [2] D. Ganji, A. Sadighi, Application of He's homotopy-perturbation method to nonlinear coupled systems of reaction-diffusion equations, *Int. J. Nonlinear Sci. Numer.* 7 (2006) 411–418.
- [3] J. Shi, P. Zhang, D. Xiao, Q. Niu, Proper definition of spin current in spin-orbit coupled systems, *Phys. Rev. Lett.* 96 (2006) 076604.
- [4] Z. Low, R. Chinga, R. Tseng, J. Lin, Design and test of a high-power high-efficiency loosely coupled planar wireless power transfer system, *IEEE Trans. Ind. Electron.* 56 (2009) 1801–1812.
- [5] M. Liu, C. Bai, Optimal harvesting of a stochastic delay competitive model, *Discrete Contin. Dyn. Syst. Ser. B* 22 (2017) 1493–1508.
- [6] E. Lakatta, V. Maltsev, T. Vinogradova, A coupled system of intracellular Ca^{2+} clocks and surface membrane voltage clocks controls the timekeeping mechanism of the heart's pacemaker, *Circ. Res.* 106 (2010) 659–673.
- [7] M. Liu, C. Bai, Analysis of a stochastic tri-trophic food-chain model with harvesting, *J. Math. Biol.* 73 (2016) 597–625.
- [8] A. Lovera, B. Gallinet, P. Nordlander, O. Martin, Mechanisms of Fano resonances in coupled plasmonic systems, *ACS Nano* 7 (2013) 4527–4536.
- [9] Z. Odibat, A note on phase synchronization in coupled chaotic fractional order systems, *Nonlinear Anal.-Real* 13 (2012) 779–789.
- [10] M. Liu, M. Fan, Stability in distribution of a three-species stochastic cascade predator-prey system with time delays, *IMA J. Appl. Math.* (2016). <http://dx.doi.org/10.1093/imamat/hxw057>.
- [11] Y. Kao, C. Wang, Global stability analysis for stochastic coupled reaction-diffusion systems on networks, *Nonlinear Anal.-Real* 14 (2013) 1457–1465.
- [12] M. Liu, C. Bai, Optimal harvesting of a stochastic mutualism model with Levy jumps, *Appl. Math. Comput.* 276 (2012) 301–309.
- [13] W. Lu, T. Chen, New approach to synchronization analysis of linearly coupled ordinary differential systems, *Physica D* 213 (2016) 214–230.
- [14] F. Wang, J. Liang, Z. Wang, F. Alsaadi, Robust synchronization of complex networks with uncertain couplings and incomplete information, *Int. J. Gen. Syst.* 45 (2016) 589–603.
- [15] Y. Liu, Z. Wang, J. Liang, X. Liu, Synchronization of coupled neutral-type neural networks with jumping-mode-dependent discrete and unbounded distributed delays, *IEEE Trans. Cybern.* 43 (2013) 102–114.
- [16] J. Li, W. Bao, S. Li, C. Wen, L. Li, Exponential synchronization of discrete-time mixed delay neural networks with actuator constraints and stochastic missing data, *Neurocomputing* 207 (2016) 700–707.
- [17] H. Song, D. Chen, W. Li, Y. Qu, Graph-theoretic approach to exponential synchronization of stochastic reaction-diffusion Cohen-Grossberg neural networks with time-varying delays, *Neurocomputing* 177 (2016) 179–187.
- [18] H. Liu, G. Hu, Y. Zhang, J. Xiao, Frequency-dependent stochastic resonance in globally coupled systems, *Commun. Theor. Phys.* 33 (2000) 197–204.
- [19] Y. Qin, J. Ren, Global existence, asymptotic behavior, and uniform attractor for a nonautonomous equation, *Math. Methods Appl. Sci.* 36 (2013) 2540–2553.
- [20] M. Liu, M. Fan, Permanence of stochastic Lotka-Volterra systems, *J. Nonlinear Sci.* 27 (2017) 425–452.
- [21] J. Lu, J. Kurths, J. Cao, N. Mahdavi, C. Huang, Synchronization control for nonlinear stochastic dynamical networks: pinning impulsive strategy, *IEEE Trans. Neural Netw. Learn.* 23 (2012) 285–292.
- [22] X. Hu, J. Lu, W. Wang, Z. Ye, Traffic signal synchronization in the saturated high-density grid road network, *Comput. Intell. Neurosci.* (2012) 532960.
- [23] M. Karimi-Ghartemani, M. Irvani, A method for synchronization of power electronic converters in polluted and variable-frequency environments, *IEEE Trans. Power Syst.* 19 (2004) 1263–1270.
- [24] J. Pursley, M.O. Mee, M. Wiltbank, Synchronization of ovulation in dairy-cows using PGF (2-ALPHA), and GNRH, *Theriogenology* 44 (1995) 915–923.
- [25] Y. Li, S. Tong, T. Li, Adaptive fuzzy backstepping control design for a class of pure-feedback switched nonlinear systems, *Nonlinear Anal.-Hybri.* 16 (2015) 72–80.

- [26] S. Dai, C. Wang, M. Wang, Dynamic learning from adaptive neural network control of a class of nonaffine nonlinear systems, *IEEE Trans. Neural Netw. Learn.* 25 (2014) 111–123.
- [27] S. Ansari, S. Das, Projective synchronization of time-delayed chaotic systems with unknown parameters using adaptive control method, *Math. Methods Appl. Sci.* 38 (2015) 726–737.
- [28] F. Wang, N. Jin, D. Liu, Q. Wei, Adaptive dynamic programming for finite-horizon optimal control of discrete-time nonlinear systems with epsilon-error bound, *IEEE Trans. Neural Netw.* 22 (2011) 24–36.
- [29] M. Kesri, On the class of infinite horizon optimal control problems affine, which are structurally stable, *Math. Methods Appl. Sci.* 39 (2016) 2799–2807.
- [30] H. Yau, C. Shieh, Chaos synchronization using fuzzy logic controller, *Nonlinear Anal.-Real* 9 (2008) 1800–1810.
- [31] C. Ting, Y. Chang, Observer-based backstepping control of linear stepping motor, *Control Eng. Pract.* 21 (2013) 930–939.
- [32] R. Yang, P. Shi, G. Liu, H. Gao, Network-based feedback control for systems with mixed delays based on quantization and dropout compensation, *Automatica* 47 (2011) 2805–2809.
- [33] H. Dong, Z. Wang, D.W.C. Ho, H. Gao, Robust H-infinity fuzzy output-feedback control with multiple probabilistic delays and multiple missing measurements, *IEEE Trans. Fuzzy Syst.* 18 (2010) 712–725.
- [34] G. Nair, F. Fagnani, S. Zampieri, R. Evans, Feedback control under data rate constraints: an overview, *Proc. IEEE* 95 (2007) 108–137.
- [35] J. Liu, J. Lian, Y. Zhuang, Output feedback L-1 finite-time control of switched positive delayed systems with MDADT, *Nonlinear Anal.-Hybri.* 15 (2015) 11–22.
- [36] S. You, W. Liu, J. Lu, X. Mao, Q. Qiu, Stabilization of hybrid systems by feedback control based on discrete-time state observations, *SIAM J. Control Optim.* 53 (2015) 905–925.
- [37] X. Mao, W. Liu, L. Hu, Q. Luo, J. Lu, Stabilization of hybrid stochastic differential equations by feedback control based on discrete-time state observations, *Systems Control Lett.* 73 (2014) 88–95.
- [38] G. Pages, H. Pham, Optimal quantization methods for nonlinear filtering with discrete-time observations, *Bernoulli* 11 (2015) 893–932.
- [39] Z. Wu, P. Shi, H. Su, J. Chu, Sampled-data exponential synchronization of complex dynamical networks with time-varying coupling delay, *IEEE Trans. Neural Netw. Learn. Syst.* 24 (2013) 1177–1187.
- [40] J. Lu, Z. Wang, J. Cao, D.W.C. Ho, J. Kurths, Pinning impulsive stabilization of nonlinear dynamical networks with time-varying delay, *Int. J. Bifurcation Chaos* (2012) 1250176.
- [41] J. Lu, D. Ho, L. Wu, Exponential stabilization of switched stochastic dynamical networks, *Nonlinearity* 22 (2009) 889–911.
- [42] Y. Yang, W. Zhou, P. Shi, X. Yang, X. Zhou, H. Su, Synchronization of delayed neural networks with Levy noise and Markovian switching via sampled data, *Nonlinear Dynam.* 81 (2015) 1179–1189.
- [43] X. Mao, *Stochastic Differential Equations and their Applications*, Horwood Publishing, Chichester, 1997.
- [44] G. Filatrell, A. Nielsen, N. Pedersen, Analysis of a power grid using a Kuramoto-like model, *Eur. Phys. J. B* 61 (2008) 485–491.
- [45] T. Peron, P. Ji, F. Rodrigues, J. Kurths, Effects of assortative mixing in the second-order Kuramoto model, *Phys. Rev. E* 91 (2015) 052805.